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Integrated hydrogeochemical and microbiological assessment of groundwater quality in a fractured volcanic aquifer, Ibb City, Yemen

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Abstract

Groundwater quality assessment in fractured volcanic aquifers presents significant challenges due to structural heterogeneity that may influence groundwater flow and contaminant transport. This study presents an integrated hydrogeochemical and microbiological assessment of groundwater from twenty production wells tapping the fractured volcanic aquifer of the Tertiary Yemen Volcanic Group in Ibb City, Yemen. Groundwater samples were collected during the dry season (November–December 2025). Hydrochemical characterization identified the Ca–Mg–HCO₃ facies as the dominant groundwater type, consistent with recharge-driven water–rock interaction and predominantly weathering of Ca- and Mg-bearing volcanic silicates, with carbonate-bearing phases representing a possible but unresolved contribution. PHREEQC saturation index modeling indicated generally near-equilibrium to oversaturated conditions with respect to carbonate minerals (calcite SI = 0.12–1.17) and persistent undersaturation with respect to halite, gypsum, and anhydrite, indicating limited thermodynamic control by these evaporite and sulfate mineral phases. Water Quality Index (WQI) values (38–95) classified all groundwater samples as excellent to good for drinking purposes, whereas irrigation indices indicated generally low sodium hazard (mean SAR = 1.59). However, fecal-coliform indicator detection occurred in 25% of the sampled wells (1–5 CFU/100 mL), indicating localized sanitary concerns. Mann–Whitney U tests showed no statistically detectable differences ($p > 0.05$) in the investigated hydrochemical parameters between fecal-coliform-positive and fecal-coliform-negative wells. Given the small number of fecal-coliform-positive wells ($n = 5$), these comparisons were considered exploratory and were not statistically powered to establish independence or equivalence between fecal-coliform indicator detection and groundwater physicochemical characteristics. These findings demonstrate that physicochemical compliance alone cannot ensure drinking-water safety in fractured volcanic aquifers and highlight the need for integrated groundwater quality assessment frameworks that combine hydrochemical and microbiological monitoring.



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Keywords Fractured volcanic aquifer, Hydrogeochemistry, Groundwater quality, Water quality index, Mineral saturation modeling, Fecal-coliform indicators

1 Introduction

Groundwater constitutes the primary source of drinking and irrigation water across Yemen, particularly in highland regions where perennial surface water resources are scarce and natural storage capacity is limited. Globally, groundwater resources are increasingly recognized as essential components of water security, especially in arid and semi-arid regions where alternative freshwater sources are constrained [32]. In Yemen, rapid urbanization, population growth, and intensified groundwater abstraction have placed increasing pressure on structurally controlled aquifer systems that historically developed under relatively low anthropogenic stress [1, 27]. Under such conditions, groundwater sustainability depends not only on resource availability but also on maintaining both chemical integrity and microbiological safety.

In volcanic aquifers, groundwater chemistry is primarily controlled by lithology and water–rock interaction processes. CO₂-driven silicate weathering, carbonate dissolution, and cation exchange commonly produce bicarbonate-dominated hydrochemical facies such as Ca–Mg–HCO₃ and Na–HCO₃ water types [16, 25, 38, 42]. These geogenic processes frequently generate groundwater that meets physicochemical drinking-water standards, particularly in recharge-dominated systems characterized by moderate residence times. However, chemical compliance alone does not necessarily guarantee sanitary protection.

Fractured volcanic aquifers represent structurally heterogeneous hydrogeological systems in which groundwater occurrence and movement are influenced by secondary porosity associated with cooling joints, tectonic fractures, and fault-related discontinuities. Such structural features can enhance groundwater recharge and create preferential flow pathways, potentially influencing the spatial distribution of dissolved constituents and microbial indicators [7, 12, 33]. However, the extent to which these structural features influence contaminant transport depends on fracture connectivity, hydraulic gradients, recharge conditions, and attenuation processes, which are not directly characterized in many groundwater-quality investigations.

Recent studies have demonstrated that groundwater may comply with physicochemical drinking-water criteria while still exhibiting microbiological contamination, highlighting the importance of combining chemical and microbiological indicators in groundwater-quality assessment [26, 40]. In fractured aquifers, the presence of interconnected fractures may provide potential preferential pathways for contaminant migration; however, the occurrence of fecal indicator organisms alone does not establish a specific transport pathway or contamination source. Therefore, microbiological observations should be interpreted together with hydrochemical characteristics and the hydrogeological setting rather than being attributed directly to fracture-controlled transport.

Despite increasing recognition of the vulnerability of fractured aquifers to contamination, integrated investigations combining hydrogeochemical characterization, geochemical modeling, irrigation suitability assessment, and microbiological analysis remain limited in volcanic highland environments of the Arabian Peninsula. In Yemen, previous studies have largely focused on basin-scale hydrochemical characterization or isolated

physicochemical analyses, with comparatively limited integration of hydrochemical processes and microbiological indicators within fractured volcanic aquifer systems.

Ibb City, located within the Tertiary Yemen Volcanic Group of the central Yemeni highlands, is underlain by a fractured basaltic aquifer that constitutes an important source of domestic water supply. Rapid urban expansion and dependence on groundwater increase the importance of evaluating both physicochemical and microbiological water quality. The present study therefore evaluates groundwater vulnerability through an integrated assessment of hydrochemical characteristics, hydrochemical evolution, mineral saturation behavior, irrigation suitability, and fecal-coliform indicator occurrence. In this framework, fecal-coliform detection is considered an indicator of localized sanitary concern, while the fractured hydrogeological setting provides important context for interpreting potential contaminant pathways. Because fracture connectivity, contaminant travel times, and specific contamination sources were not directly investigated, the study does not assume a demonstrated fracture-controlled transport mechanism.

We hypothesize that conventional physicochemical classifications may not fully capture sanitary risks in fractured volcanic aquifers because microbiological quality can vary independently of major-ion chemistry. Specifically, groundwater in Ibb City is expected to exhibit hydrochemical characteristics consistent with recharge-related water–rock interaction while showing localized fecal-coliform indicator detection that may not correspond to differences in bulk physicochemical quality.

To test this hypothesis, this study presents an integrated hydrogeochemical and microbiological assessment of groundwater from representative production wells in Ibb City. The analytical framework combines major-ion hydrochemistry, hydrochemical facies analysis, geochemical process interpretation, thermodynamic mineral saturation modeling, irrigation suitability evaluation, and bacteriological testing based on fecal-coliform indicators. By integrating these complementary lines of evidence, the study aims to characterize groundwater quality, identify localized sanitary concerns, and provide a scientifically grounded basis for groundwater-quality monitoring, wellhead protection, and sustainable groundwater management in urbanizing volcanic aquifers of Yemen.

2 Materials and methods

2.1 Study area and hydrogeological setting

The study area is located in Ibb City, central Yemen, between latitudes 13°33′–14°24′ N and longitudes 43°40′–44°40′ E (Fig. 1). The region forms part of the southern-central Yemeni highlands and is characterized by rugged volcanic topography with elevations ranging from approximately 1400 to 3200 m above mean sea level. The pronounced relief influences surface runoff distribution, hydraulic gradients, and spatial recharge patterns within structurally controlled zones. Climatically, the area represents a transitional regime between semi-humid and semi-arid conditions influenced by high-altitude orographic effects. Mean annual precipitation ranges from 700 to 1000 mm, concentrated during the monsoon season (April–August), whereas the dry season extends from October to March [28]. Mean seasonal air temperatures average approximately 18 °C in winter and 24 °C in summer [43], supporting seasonally controlled meteoric recharge to the volcanic aquifer system.

Ibb City represents one of the most densely populated urban centers in the Yemeni highlands, with an estimated population exceeding 300,000 inhabitants. Rapid urban

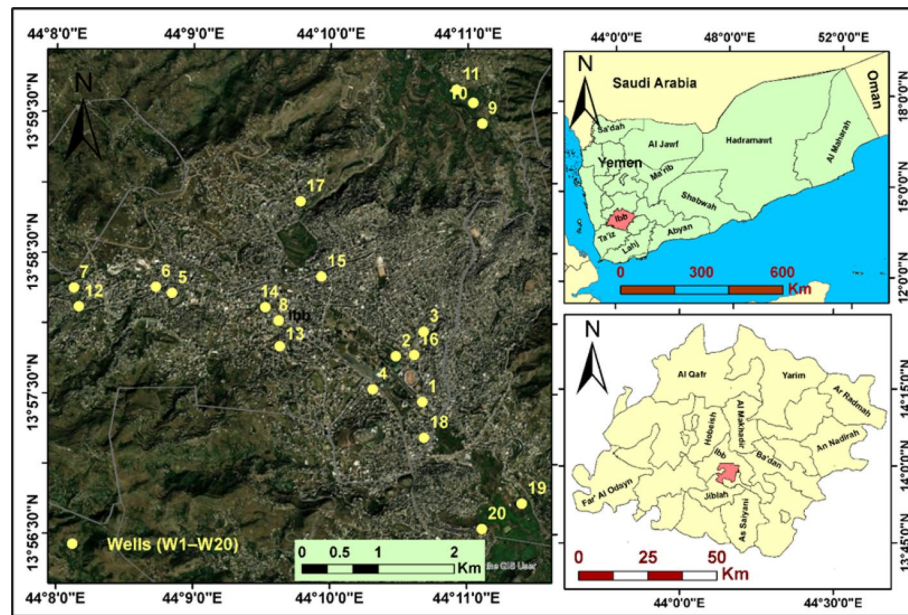


Fig. 1 Location of the study area and distribution of groundwater sampling wells in Ibb City, Yemen

expansion and population growth have increased pressure on groundwater resources and sanitation infrastructure, potentially enhancing the vulnerability of the fractured volcanic aquifer to contamination from septic systems and wastewater leakage [13].

Geologically, the area is underlain by the Tertiary Yemen Volcanic Group, composed predominantly of stratified basaltic lava flows interbedded with ignimbrites, pyroclastics, and trachytic units [8]. These formations constitute the principal aquifer host rocks and define the mineralogical framework controlling groundwater chemistry. Hydrogeologically, groundwater occurs within a fractured volcanic aquifer system in which storage and flow are controlled by secondary porosity associated with cooling joints, tectonic fractures, and fault-related discontinuities. This fracture-dominated architecture produces heterogeneous hydraulic conductivity and promotes preferential flow relative to granular porous media, resulting in relatively rapid groundwater circulation and increased susceptibility to structurally mediated recharge and contaminant transport. Fracture-controlled flow in hard-rock aquifers commonly produces strong hydraulic heterogeneity and preferential pathways that can accelerate recharge and contaminant transport compared with granular porous media [7, 33].

Groundwater recharge occurs predominantly through direct infiltration of seasonal rainfall along weathered basalt horizons, tectonic fractures, and fault-related discontinuities. Municipal production wells tap the fractured basaltic aquifer, which constitutes the principal source of domestic water supply in Ibb City.

2.2 Groundwater sampling and field measurements

Twenty groundwater samples were collected from representative municipal production wells (W1–W20) across Ibb City (Fig. 1). Sampling was conducted during the dry season (November–December 2025) to minimize short-term recharge effects and characterize baseline hydrochemical conditions. Prior to sampling, wells were purged for 15–20 min until stabilization of pH and EC (variation < 5%) to ensure collection of representative groundwater [6]. Field parameters (temperature, pH, electrical conductivity, and

total dissolved solids) were measured in situ using a calibrated portable multiparameter meter, with daily calibration using certified standards. Samples for chemical analysis were collected in pre-cleaned high-density polyethylene (HDPE) bottles and filtered through 0.45 μm membranes. Cation samples were acidified with ultrapure HNO_3 to $\text{pH} < 2$, whereas anion samples were not acidified. Microbiological samples were collected aseptically in sterile glass bottles, stored at approximately 4 $^{\circ}\text{C}$, and transported to the laboratory within six hours. All procedures followed APHA [2] standard sampling and preservation protocols.

2.3 Laboratory analysis and quality control

Chemical analyses were conducted at the laboratories of the Ibb Water and Sanitation Local Corporation in accordance with the *Standard Methods for the Examination of Water and Wastewater* [2]. Major cations (Ca^{2+} , Mg^{2+} , Na^{+} , and K^{+}) and anions (HCO_3^{-} , Cl^{-} , SO_4^{2-} , NO_3^{-} , and F^{-}) were determined using standard analytical procedures. Ca^{2+} and Mg^{2+} were determined by EDTA titration, Cl^{-} by argentometric titration using AgNO_3 , alkalinity (reported as HCO_3^{-}) by acid titration with standardized H_2SO_4 , Na^{+} and K^{+} by flame photometry, and SO_4^{2-} , NO_3^{-} , and F^{-} by spectrophotometric methods. Quality assurance and quality control (QA/QC) included reagent blanks, duplicate samples, calibration standards, and replicate measurements.

Internal consistency of the hydrochemical data was evaluated using the ionic charge-balance error (CBE), calculated after converting ionic concentrations to milliequivalents per liter (meq/L):

$$\text{CBE (\%)} = \left[\frac{(\sum \text{Cations} - \sum \text{Anions})}{(\sum \text{Cations} + \sum \text{Anions})} \right] \times 100 \quad (1)$$

The cation sum included Na^{+} , Ca^{2+} , Mg^{2+} , and K^{+} , whereas the anion sum included Cl^{-} , HCO_3^{-} , SO_4^{2-} , NO_3^{-} , and F^{-} . The CBE was calculated separately for each groundwater sample. An absolute CBE of $\leq 5\%$ was used as the primary analytical quality-control criterion. Eighteen of the twenty samples satisfied this criterion, whereas W15 (-5.17%) and W20 ($+5.85\%$) showed marginal deviations. These two samples were retained because their deviations were relatively small, and a sensitivity check excluding W15 and W20 resulted in only a minor change in the mean absolute CBE, from 3.05 to 2.77%, indicating that their inclusion did not materially alter the overall charge-balance characteristics of the dataset. Individual CBE values are provided in Supplementary Table S1.

2.4 Microbiological analysis

Fecal coliforms (FC) were determined using the membrane filtration technique [2]. A 100 mL aliquot of each groundwater sample was filtered through a sterile 0.45 μm membrane, placed on m-FC agar, and incubated at 44.5 ± 0.2 $^{\circ}\text{C}$ for 24 h. Characteristic colonies were enumerated and expressed as colony-forming units per 100 mL (CFU/100 mL). Quality assurance procedures included sterile blanks and duplicate analyses. No confirmatory test specifically targeting *Escherichia coli* or other specific pathogenic microorganisms was performed. Therefore, the microbiological results are interpreted as fecal-coliform indicator detection and not as confirmation of *E. coli* or specific pathogen presence.

2.5 Hydrogeochemical interpretation and data processing

Hydrogeochemical characterization was conducted using an integrated analytical framework combining graphical methods, ionic relationships, thermodynamic modeling, and water quality indices. All ionic concentrations were converted from mg/L to milliequivalents per liter (meq/L) prior to graphical representation and index calculations to ensure stoichiometric consistency.

2.5.1 Hydrochemical facies classification

Hydrochemical facies were determined using Piper trilinear diagrams to classify groundwater types based on the relative proportions of major cations (Ca^{2+} , Mg^{2+} , $\text{Na}^+ + \text{K}^+$) and anions (HCO_3^- , SO_4^{2-} , Cl^-). This approach facilitates identification of dominant water types and hydrochemical evolution trends within the aquifer system. Durov diagrams were additionally employed to corroborate facies classification and to evaluate geochemical processes including mixing, ion exchange, and mineral dissolution. Hydrochemical facies classification provides an effective basis for linking groundwater types to dominant geochemical processes and flow evolution in heterogeneous aquifers [35].

2.5.2 Mechanisms controlling groundwater chemistry

Gibbs diagrams were applied to assess the primary mechanisms controlling groundwater composition. The ratios $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ and $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$ were plotted against total dissolved solids (TDS) to distinguish among precipitation dominance, rock–water interaction, and evaporation–crystallization processes. Complementary bivariate ionic relationships were analyzed to evaluate mineral dissolution patterns and potential cation-exchange processes within the fractured volcanic aquifer. Additional diagnostic relationships were used to evaluate sodium–chloride behavior and ion-exchange processes. The Na^+ versus Cl^- relationship was evaluated using concentrations expressed in milliequivalents per liter (meq/L), with the 1:1 line used as a reference for assessing the relative behavior of Na^+ and Cl^- . The Chloro-Alkaline Indices (CAI-I and CAI-II) were calculated as follows:

$$\text{CAI} - \text{I} = \frac{\text{Cl}^- - (\text{Na}^+ + \text{K}^+)}{\text{Cl}^-} \quad (2)$$

$$\text{CAI} - \text{II} = \frac{\text{Cl}^- - (\text{Na}^+ + \text{K}^+)}{\text{HCO}_3^- + \text{SO}_4^{2-}} \quad (3)$$

where all ionic concentrations are expressed in meq/L. Positive CAI values are interpreted as consistent with reverse cation exchange, whereas negative values are consistent with direct cation exchange. The CAI-I and CAI-II values were calculated for all twenty groundwater samples and used as complementary indicators of spatially variable ion-exchange behavior [10].

2.5.3 Saturation index modeling

Thermodynamic equilibrium between groundwater and selected mineral phases was evaluated using PHREEQC Interactive (version 3.8.6-17100) [29]. Calculations were conducted using the phreeqc.dat thermodynamic database with the Davies activity-coefficient correction. Saturation indices (SI) were calculated for selected carbonate, sulfate, and evaporite minerals to evaluate mineral stability and water–rock interaction. For

each groundwater sample, the measured temperature and pH were entered directly into the PHREEQC SOLUTION definition, and all dissolved constituent concentrations were specified in mg/L. Bicarbonate was entered explicitly as alkalinity using the PHREEQC syntax Alkalinity ... as HCO₃. The major ions included in the PHREEQC input were Na⁺, Ca²⁺, Mg²⁺, K⁺, Cl⁻, HCO₃⁻, SO₄²⁻, and NO₃⁻. No redox constraint was imposed because Eh/pe was not measured during field sampling.

No analytical concentration was adjusted, charge-balanced, or otherwise modified before PHREEQC speciation. In particular, no balancing ion or charge-balance adjustment was applied within the PHREEQC SOLUTION definitions. The final laboratory dataset, including the corrected concentrations described in Sect. 2.3, was used directly for the saturation-index calculations.

The saturation index was calculated as:

$$SI = \log\left(\frac{IAP}{K_{sp}}\right) \quad (4)$$

where IAP denotes the ion activity product and K_{sp} represents the solubility product constant. An SI ≈ 0 indicates equilibrium, SI > 0 indicates supersaturation (precipitation tendency), and SI < 0 indicates undersaturation (dissolution tendency).

A complete PHREEQC input file containing the SOLUTION definitions for all twenty groundwater samples (W1–W20), together with the model settings used to generate the selected output, is provided as Supplementary Material. The PHREEQC input file documents the sample-specific temperature, pH, analytical concentrations, alkalinity specification, and selected-output configuration used to generate the saturation indices reported in Table 4.

2.6 Water quality assessment indices

2.6.1 Drinking water quality index (WQI)

The WQI was calculated using eleven physicochemical parameters: pH, TDS, Na⁺, Ca²⁺, Mg²⁺, K⁺, Cl⁻, HCO₃⁻, SO₄²⁻, NO₃⁻, and F⁻. Following the weighted relative-importance approach used in groundwater-quality assessment, each parameter was assigned a weight (*w_i*) ranging from 1 to 5 according to its relative importance to drinking-water quality and potential implications for human health [34]. Recent groundwater-quality assessments have similarly combined hydrochemical characterization with water quality

Table 1 WHO reference values (*S_i*), assigned weights (*w_i*), and relative weights (*W_i*) used for Water Quality Index (WQI) calculation

Chemical parameter	WHO guideline value (<i>S_i</i>)	Assigned weight (<i>w_i</i>)	Relative weight (<i>W_i</i>)
pH	8.5	3	0.0789
TDS	1000	5	0.1316
Na ⁺	200	3	0.0789
Ca ²⁺	75	2	0.0526
Mg ²⁺	50	3	0.0789
K ⁺	12	2	0.0526
Cl ⁻	250	4	0.1053
HCO ₃ ⁻	150	5	0.1316
SO ₄ ²⁻	250	3	0.0789
NO ₃ ⁻	50	4	0.1053
F ⁻	1.5	4	0.1053
Total	–	Σ <i>w_i</i> = 38	Σ <i>W_i</i> = 1.000

indices to evaluate drinking-water suitability and groundwater mineralization [4, 20, 23]. Higher weights were assigned to parameters considered more influential within the adopted assessment framework, whereas lower weights were assigned to parameters of comparatively lower relative importance. The assigned weights (w_i), WHO reference values (S_i), and calculated relative weights (W_i) are presented in Table 1.

The relative weight (W_i) for each parameter was calculated as:

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (5)$$

where W_i is the relative weight of parameter i , w_i is the assigned weight of parameter i , and n is the total number of parameters included in the WQI calculation.

The quality rating (Q_i) for each parameter was computed using:

$$Q_i = \frac{Ci - C_{i0}}{S_i - C_{i0}} \times 100 \quad (6)$$

where Q_i is the quality rating of parameter i , C_i is the measured concentration of parameter i , C_{i0} is the ideal value of parameter i , and S_i is the corresponding guideline value.

The ideal value (C_{i0}) was set to zero for all parameters except pH, for which an ideal value of 7.0 was adopted [37].

The sub-index (SI_i) was calculated as:

$$SI_i = W_i \times Q_i \quad (7)$$

where SI_i is the sub-index of parameter i , W_i is the relative weight assigned to parameter i , and Q_i is the quality rating of parameter i .

The overall WQI for each sample was calculated as the sum of the individual sub-indices:

$$WQI = \sum_{i=1}^n SI_i \quad (8)$$

where WQI is the overall Water Quality Index, SI_i is the sub-index for parameter i , and n is the total number of parameters included in the calculation.

A complete worked example for groundwater sample W1 is provided in Supplementary Table S2.

Groundwater quality was classified into five categories: excellent (< 50), good (50–100), poor (100–200), very poor (200–300), and unsuitable for drinking (> 300), following the classification proposed by Brown et al. [11].

2.6.2 Irrigation water quality assessment

Irrigation suitability interpretation was additionally guided by FAO recommendations that integrate salinity and sodicity hazards with soil and crop sensitivity considerations [5]. All ionic concentrations were converted from mg/L to milliequivalents per liter (meq/L) prior to calculation.

Electrical conductivity was used as an indicator of salinity hazard, reflecting total dissolved salts and their influence on soil osmotic pressure and plant water uptake. Salinity classes were interpreted according to the U.S. Salinity Laboratory (USSL) classification [31].

The sodium hazard was assessed using the Sodium Adsorption Ratio (SAR), calculated as:

$$SAR = Na^+ / \sqrt{[(Ca^{2+} + Mg^{2+})/2]} \quad (9)$$

where Na^+ , Ca^{2+} , and Mg^{2+} represent sodium, calcium, and magnesium concentrations, respectively, expressed in meq/L. SAR values were classified following USSL criteria.

Sodium percentage (Na%) was calculated to evaluate the proportion of sodium relative to total major cations:

$$Na\% = [(Na^+ + K^+) / (Ca^{2+} + Mg^{2+} + Na^+ + K^+)] \times 100 \quad (10)$$

where Na^+ , K^+ , Ca^{2+} , and Mg^{2+} represent the concentrations of sodium, potassium, calcium, and magnesium, respectively, expressed in meq/L. Na% values were classified according to the Wilcox [41] irrigation suitability criteria.

Residual Sodium Carbonate (RSC) was calculated to assess the potential effect of bicarbonate on calcium and magnesium precipitation and soil sodicity:

$$RSC = HCO_3^- (Ca^{2+} + Mg^{2+}) \quad (11)$$

where RSC is the Residual Sodium Carbonate expressed in meq/L, and HCO_3^- , Ca^{2+} , and Mg^{2+} represent the concentrations of bicarbonate, calcium, and magnesium, respectively, expressed in meq/L.

RSC values were interpreted using standard threshold limits [31].

Combined salinity and sodium hazards were visualized using the USSL diagram (EC vs. SAR) and the Wilcox diagram (Na% vs. EC) to provide an integrated assessment of irrigation suitability.

2.7 Statistical analysis

Descriptive statistics were computed for all measured variables to summarize central tendency and dispersion. Pearson's correlation coefficients (r) were calculated to evaluate linear relationships among hydrochemical parameters and to support interpretation of dominant geochemical processes.

Groundwater samples were grouped into fecal coliform-positive and fecal coliform-negative wells. Two-sided non-parametric Mann-Whitney U tests were applied to evaluate potential differences between FC-positive and FC-negative wells for WQI, EC, TDS, and NO_3^- . The asymptotic two-sided p -values were calculated with continuity correction and appropriate tie correction. For each comparison, Mann-Whitney U statistics, group medians, and rank-biserial correlation coefficients were reported to quantify the magnitude and direction of between-group differences. The rank-biserial correlation was calculated as:

$$r_{r\beta} = 1 - \frac{2U}{n_1 n_2} \quad (12)$$

where $r_{r\beta}$ is the rank-biserial correlation coefficient, U is the Mann-Whitney U statistic for the FC-positive group, n_1 is the number of FC-positive wells, and n_2 is the number of FC-negative wells. Positive values indicate higher ranks in the FC-positive group, whereas negative values indicate higher ranks in the FC-negative group.

Given the small number of fecal-coliform-positive wells ($n = 5$), the analysis was considered exploratory and was not powered to establish statistical independence or equivalence between fecal-coliform indicator detection and the investigated physicochemical variables.

Statistical significance was considered at $p < 0.05$ and $p < 0.01$. All statistical analyses were performed using IBM SPSS Statistics (version 26). Hydrochemical diagrams (Piper, Durov, Gibbs, USSSL, and Wilcox) were generated using RockWorks (version 17) and Grapher software, whereas mineral saturation indices were computed using PHREEQC Interactive (version 3.8.6-17100). Given the limited sample size ($n = 20$), multivariate statistical techniques such as principal component analysis (PCA) were not considered statistically robust. Accordingly, interpretation was based on correlation analysis in conjunction with process-oriented hydrogeochemical diagnostics.

3 Results and discussion

3.1 Physicochemical and microbiological characteristics

The physicochemical and microbiological characteristics of groundwater from the twenty production wells (W1–W20) are summarized in Table 2 and interpreted in relation to World Health Organization [40] drinking-water guidelines.

3.1.1 Physicochemical characteristics

Groundwater temperature ranged from 21.1 to 25.0 °C (mean: 23.7 °C), reflecting stable dry-season subsurface conditions in the highland volcanic setting. pH values (7.1–8.4; mean: 7.7) indicate slightly alkaline conditions consistent with carbonate buffering within basaltic terrains. Electrical Conductivity (EC) ranged from 488 to 1,406 $\mu\text{S}/\text{cm}$ (mean: 966 $\mu\text{S}/\text{cm}$), and Total Dissolved Solids (TDS) ranged from 317 to 914 mg/L (mean: 628 mg/L). All samples remained below 1000 mg/L, which is commonly considered acceptable for palatability/acceptability rather than a strict health-based limit.

Elevated mineralization observed in wells W17 and W19 suggests intensified water–rock interaction along structurally controlled flow paths rather than evaporative concentration. The dominant cation and anion sequences are consistent with water–rock interaction involving weathering of Ca- and Mg-bearing volcanic silicates, while carbonate-bearing phases may provide an additional but unresolved contribution. Total Hardness (TH) ranged from 46 to 638 mg/L as CaCO_3 (mean: 377 mg/L). Four wells (W7, W8, W13, and W17) exceeded 500 mg/L, classifying them as very hard water. Elevated hardness reflects dissolution of Ca- and Mg-bearing minerals within the fractured volcanic matrix.

3.1.2 Nutrient and trace ion variability

Nitrate (NO_3^-) concentrations ranged from 0.1 to 60 mg/L (mean: 13.3 mg/L). While most samples complied with the 50 mg/L regulatory limit, well W9 exceeded this threshold, indicating localized anthropogenic nitrogen input likely associated with urban recharge sources. Fluoride (F^-) concentrations remained low (0–1.4 mg/L; mean: 0.4 mg/L), with all samples below the WHO guideline value (1.5 mg/L), suggesting limited fluorine-bearing mineral influence. Elevated nitrate in urban and peri-urban groundwater is commonly linked to sanitation leakage, septic systems, and diffuse wastewater inputs, particularly where rapid recharge and shallow flow pathways occur [26].

Table 2 Physicochemical and bacteriological characteristics of groundwater samples

Well No.	T	pH	EC	TDS	Na ⁺	Ca ²⁺	Mg ²⁺	K ⁺	Cl ⁻	HCO ₃ ⁻	SO ₄ ²⁻	NO ₃ ⁻	F ⁻	TH	Fecal coliform
W1	22.0	7.4	829	539	41	102	27	1.2	80	366	39	12.0	0.3	367	0
W2	24.3	8.0	1137	739	48	128	33	1.5	95	416	64	11.0	0.4	457	0
W3	25.0	7.7	842	547	38	96	29	1.5	82	348	42	13.0	0.3	362	0
W4	22.0	8.0	724	471	140	24	7	0.5	110	280	39	6.5	0.5	90	1
W5	24.3	7.2	977	635	55	132	16	0.8	114	325	39	6.3	0.3	397	0
W6	24.8	7.3	1102	716	70	118	32	1.8	126	415	68	20.0	0.5	427	5
W7	24.4	7.1	1270	826	23	180	22	0.8	125	427	61	5.3	0.2	541*	0
W8	25.0	7.1	1154	750	40	144	38	1.7	98	439	47	8.5	0.3	518*	5
W9	22.0	8.2	946	615	96	78	40	1.5	100	414	59	60.0*	0.7	360	0
W10	23.5	8.2	766	498	74	50	16	1.8	104	146	47	7.8	0.1	192	0
W11	22.0	8.0	878	571	85	58	35	1.3	117	224	53	34.8	0.5	292	0
W12	24.8	8.4	488	317	68	15	2	0.5	60	96	34	6.7	1.4	46	0
W13	25.0	7.2	1237	804	39	182	22	0.7	107	450	63	7.5	0.2	546*	3
W14	24.6	7.4	699	454	39	180	10	1.3	107	387	50	7.7	0.2	492	4
W15	23.5	7.8	805	523	42	80	29	0.9	72	421	7	0.1	0.9	322	0
W16	24.5	8.1	983	639	40	110	38	1.4	95	381	32	9.5	0.3	432	0
W17	21.1	7.3	1406	914	33	156	60	0.6	139	500	88	23.0	0.3	638*	0
W18	24.0	7.4	1187	772	62	112	32	1.3	123	405	61	13.0	0.4	412	0
W19	23.0	7.6	1366	888	59	122	37	1.6	98	468	82	4.3	1.0	457	0
W20	23.7	8.3	521	339	35	48	18	0.8	60	118	50	8.5	0.3	196	0
Min	21.1	7.1	488	317	23	15	2	0.5	60	96	7	0.1	0.1	46	0
Max	25.0	8.4	1406	914	140	182	60	1.8	139	500	88	60.0	1.4	638	5
Mean	23.7	7.7	966	628	56.4	106.1	26.6	1.2	101.4	341.2	51.1	13.3	0.4	377	-
SD (±)	1.3	0.4	248.4	161.4	28.3	48.2	13.5	0.4	20.8	115.8	17.5	13.5	0.3	146.4	-
WHO reference value (2022)	-	6.5–8.5	1500	1000	200	-	-	12	250	-	250	50	1.5	500	0
Yemeni drinking-water standard	-	6.5–9.0	1500	1000	200	100	50	12	250	-	250	50	1.5	500	0

All parameters are expressed in mg/L (TH as CaCO₃), except T (°C), EC (µS/cm), pH, and fecal coliforms (CFU/100 mL). WHO reference values are based on the Guidelines for Drinking-water Quality (140). Where no health-based guideline value is established, the reported WHO reference value represents an acceptability or operational consideration rather than a health-based limit. In particular, the reference values for pH, TDS, Na⁺, K⁺, Cl⁻, SO₄²⁻, and TH are primarily related to operational, acceptability, or aesthetic considerations, whereas the values for NO₃⁻ and F⁻ are health-based guideline values.

Bold values and an asterisk (*) indicate values exceeding the corresponding WHO reference value or Yemeni drinking-water standard.

3.1.3 Microbiological quality

Fecal coliforms (FC) were detected in five of the twenty sampled production wells (W4, W6, W8, W13, and W14), representing 25% of the sampled sources, with concentrations ranging from 1 to 5 CFU/100 mL. The remaining fifteen wells showed no detectable fecal coliforms. The detected FC concentrations indicate localized microbiological concerns within the sampled groundwater sources and demonstrate that microbiological quality was not uniform across the production wells.

3.2 Hydrochemical facies and groundwater evolution

Hydrochemical facies of groundwater in the Ibb volcanic aquifer were interpreted using Piper and Durov diagrams to identify dominant water types and evolutionary trends within the fractured volcanic system [14, 30]. These graphical approaches integrate major-ion chemistry and provide a robust framework for assessing lithological controls on groundwater evolution.

The Piper diagram (Fig. 2) shows that most groundwater samples cluster within the Ca–Mg–HCO₃ facies, occupying the alkaline earth–bicarbonate domain. This distribution reflects the dominance of Ca²⁺ and Mg²⁺ over Na⁺ + K⁺ and of HCO₃[−] over Cl[−] and SO₄^{2−}, indicating groundwater mineralization predominantly controlled by water–rock interaction within the volcanic aquifer. The observed Ca–Mg–HCO₃ hydrochemical facies is consistent with recharge-dominated conditions and is primarily compatible with water–rock interaction involving weathering of Ca- and Mg-bearing volcanic silicates. The contribution of carbonate-bearing phases to groundwater Ca²⁺ and HCO₃[−] cannot

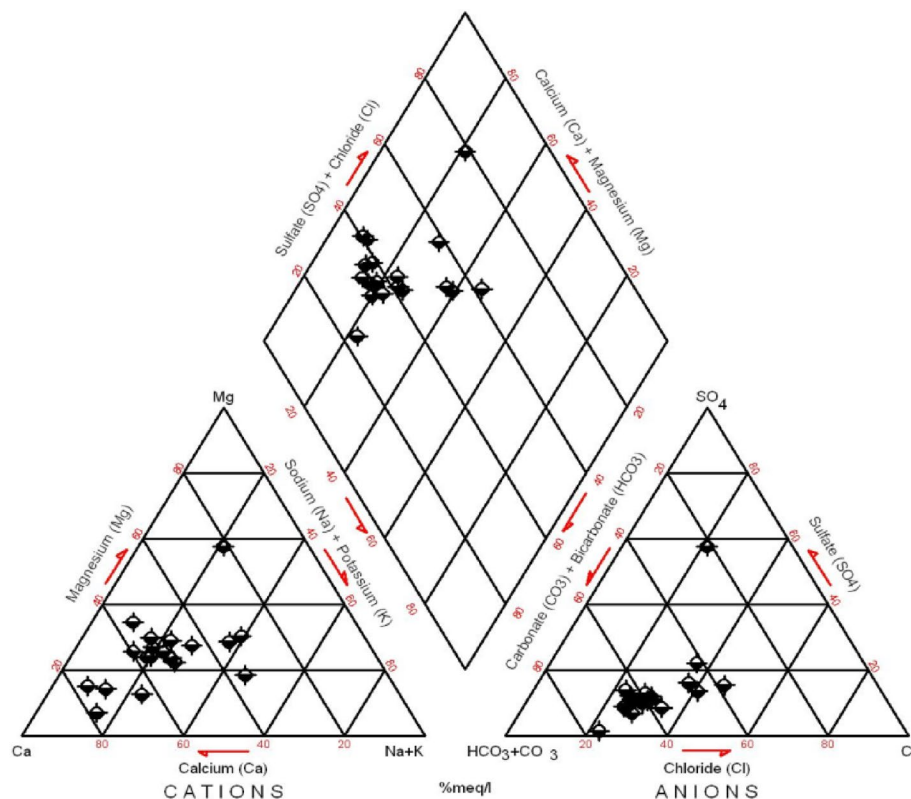


Fig. 2 Piper diagram showing hydrochemical facies classification of groundwater samples

be excluded, but their specific occurrence and contribution could not be directly established from the available mineralogical data.

Limited dispersion toward chloride- or sulfate-dominated fields suggests that evaporite dissolution, saline mixing, or widespread anthropogenic salinization are not major regional controls. Instead, groundwater chemistry appears largely controlled by geogenic processes associated with basaltic and ignimbritic lithologies of the Tertiary Yemen Volcanic Group. The clustering pattern further indicates a system that has not progressed toward advanced chemical evolution typically characterized by Na–Cl or SO_4 -dominant facies.

The Durov diagram (Fig. 3) corroborates these findings, with most samples plotting within bicarbonate-dominated fields associated with alkaline earth cations. A localized transition toward mixed Ca–Na– HCO_3 facies is observed in well W4, suggesting partial cation exchange along fracture-controlled flow paths. The spatial restriction of this transitional facies indicates structural heterogeneity and variable water–rock interaction intensity rather than basin-scale chemical transformation. Overall, the combined Piper–Durov interpretation indicates that groundwater evolution in the Ibb volcanic aquifer is predominantly controlled by lithology and water–rock interaction within the volcanic sequence. The hydrochemical evolution is primarily associated with silicate weathering

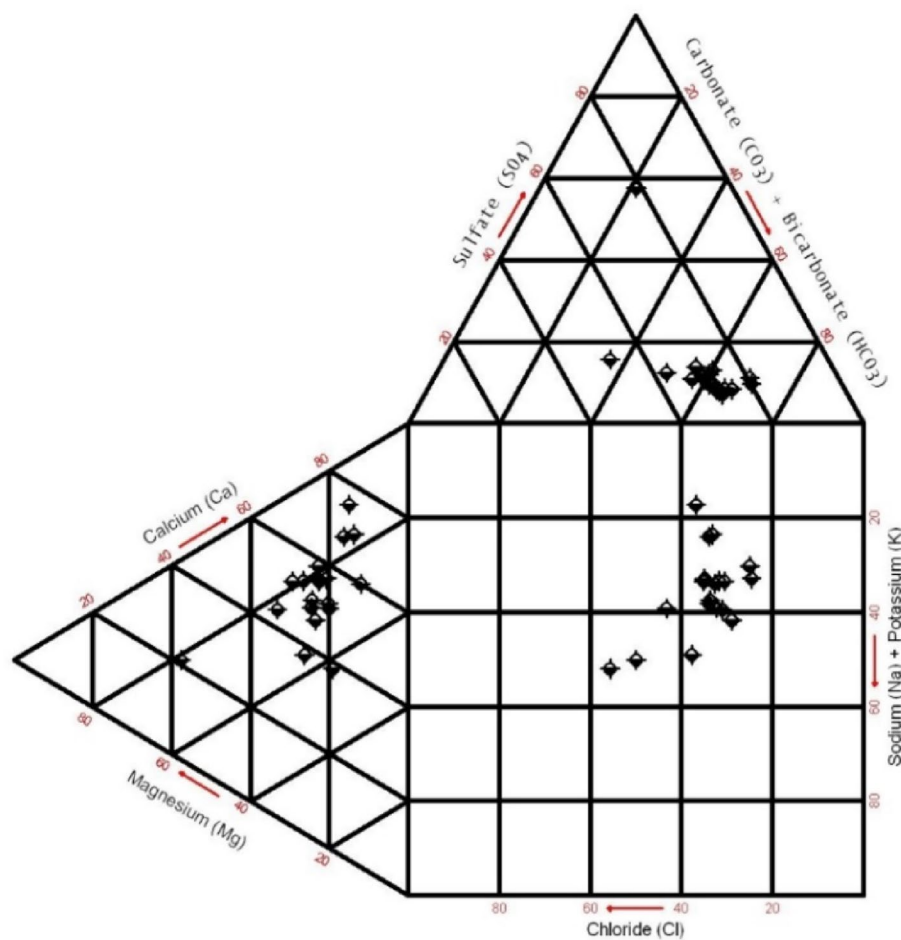


Fig. 3 Durov diagram illustrating hydrochemical processes and dominant water types

under recharge-dominated conditions, accompanied by localized dissolution of secondary carbonate phases and limited cation-exchange processes.

3.3 Mechanisms controlling groundwater chemistry

An integrated analytical framework combining Gibbs diagrams, bivariate ionic relationships, and Pearson correlation analysis was applied to elucidate the dominant geochemical processes controlling groundwater composition in the fractured volcanic aquifer of Ibb City.

3.3.1 Rock–water interaction dominance

Gibbs diagrams (Fig. 4), constructed by plotting TDS against the ionic ratios $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ and $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$, show that most groundwater samples cluster within the rock–water interaction field [19]. This distribution indicates that dissolved constituents are primarily derived from mineral dissolution and weathering reactions within the fractured basaltic and ignimbritic units of the Tertiary Yemen Volcanic Group. The absence of samples within the precipitation-dominance field suggests that rainfall chemistry does not exert primary control on groundwater composition. Likewise, limited dispersion toward the evaporation–crystallization domain indicates minor evaporative influence under prevailing high-altitude climatic conditions. Similar recharge-driven hydrochemical patterns have been reported in fractured volcanic aquifers where lithological control dominates groundwater evolution [3].

3.3.2 Carbonate dissolution and silicate weathering contributions

Hydrochemical facies of groundwater in the Ibb volcanic aquifer were interpreted using Piper and Durov diagrams to identify dominant water types and evolutionary trends within the fractured volcanic system [14, 30]. These graphical approaches integrate major-ion chemistry and provide a framework for assessing the hydrochemical evolution of groundwater. The Piper diagram (Fig. 2) shows that most groundwater samples cluster within the Ca–Mg– HCO_3 facies, occupying the alkaline earth–bicarbonate domain. This distribution reflects the dominance of Ca^{2+} and Mg^{2+} over $\text{Na}^+ + \text{K}^+$ and of HCO_3^- over Cl^- and SO_4^{2-} , indicating groundwater mineralization predominantly controlled by water–rock interaction within the volcanic aquifer. The observed Ca–Mg– HCO_3 hydrochemical facies is consistent with recharge-dominated conditions and is primarily compatible with water–rock interaction involving weathering of Ca- and Mg-bearing

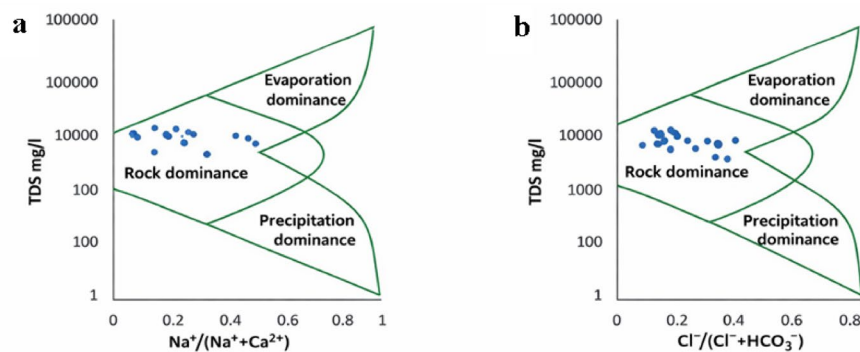


Fig. 4 Gibbs diagrams illustrating dominant geochemical processes controlling groundwater chemistry: **a** TDS versus $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ and **b** TDS versus $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$

volcanic silicates. The contribution of carbonate-bearing phases to groundwater Ca^{2+} and HCO_3^- cannot be excluded, but their specific occurrence and contribution could not be directly established from the available mineralogical data [3].

Limited dispersion toward chloride- or sulfate-dominated fields suggests that evaporite dissolution, saline mixing, or widespread anthropogenic salinization are not major regional controls. Instead, groundwater chemistry appears largely controlled by geogenic processes associated with basaltic and ignimbritic lithologies of the Tertiary Yemen Volcanic Group. The clustering pattern further indicates a system that has not progressed toward advanced chemical evolution typically characterized by Na–Cl or SO_4 -dominant facies.

To provide additional evidence regarding the processes controlling groundwater chemistry, bivariate ionic relationships were evaluated in milliequivalent units (meq/L) (Fig. 5). The plot of $(\text{Ca}^{2+} + \text{Mg}^{2+})$ versus $(\text{HCO}_3^- + \text{SO}_4^{2-})$ shows that most samples cluster near the 1:1 equiline, indicating a broad balance among the major ions involved in water–rock interaction. In the Ca^{2+} versus HCO_3^- relationship (Fig. 5c), several samples deviate toward the 2:1 trend, which is compatible with contributions from weathering of Ca-bearing volcanic minerals, such as plagioclase hydrolysis. However, these relationships alone cannot resolve the relative contributions of silicate weathering, carbonate-bearing phases, and cation-exchange processes. The SO_4^{2-} versus Ca^{2+} plot (Fig. 5d) shows a pronounced departure from the 1:1 line, indicating that sulfate is not

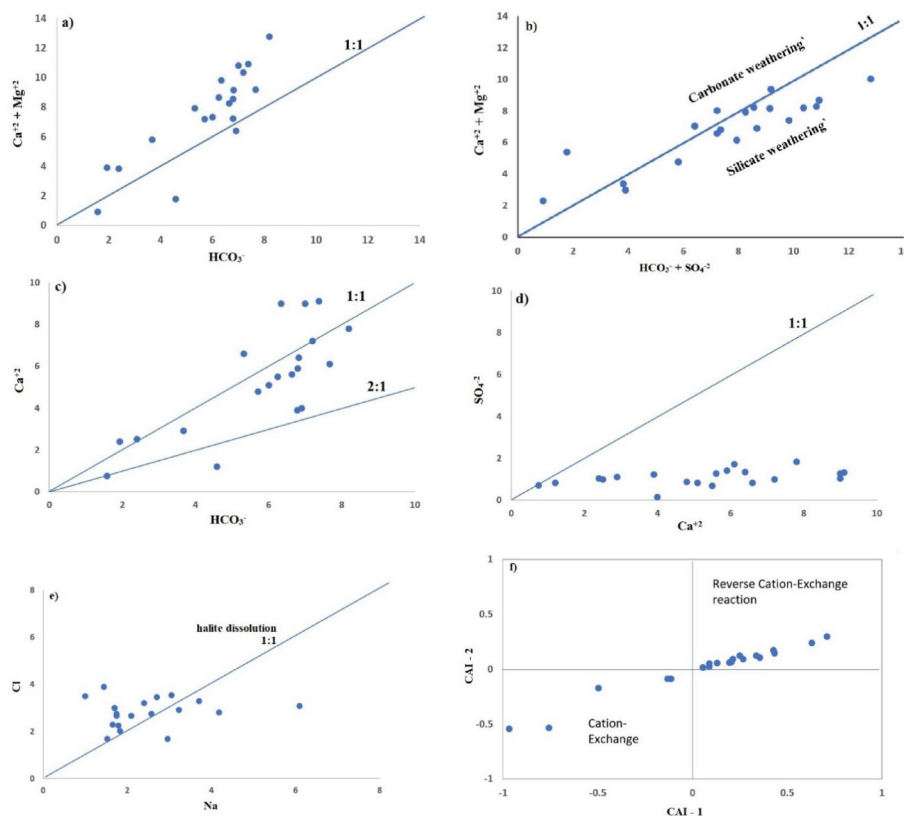


Fig. 5 Bivariate ionic relationships and ion-exchange indicators (meq/L) used to assess the geochemical processes controlling groundwater chemistry in the fractured volcanic aquifer of Ibb City: **a** $(\text{Ca}^{2+} + \text{Mg}^{2+})$ versus HCO_3^- , **b** $(\text{Ca}^{2+} + \text{Mg}^{2+})$ versus $(\text{HCO}_3^- + \text{SO}_4^{2-})$, **c** Ca^{2+} versus HCO_3^- showing the 1:1 and 2:1 reference lines, **d** SO_4^{2-} versus Ca^{2+} , **e** Na^+ versus Cl^- showing the 1:1 reference line, and **f** CAI-II versus CAI-I

dominantly controlled by gypsum or anhydrite dissolution. This interpretation is consistent with the PHREEQC results (Sect. 3.4), which indicate persistent undersaturation with respect to sulfate minerals.

Additional diagnostic relationships were used to evaluate sodium–chloride behavior and potential ion-exchange processes. The Na^+ versus Cl^- relationship (Fig. 5e) showed a weak and statistically non-significant association ($r = 0.130$, $p = 0.585$), indicating that Na^+ and Cl^- do not exhibit a simple common-source relationship across the sampled wells. Deviations from the 1:1 relationship may reflect contributions from ion exchange, silicate weathering, or other water–rock interaction processes, as reported in comparable groundwater hydrogeochemical assessments [9, 10, 36]. The Chloro-Alkaline Indices (CAI-I and CAI-II) were calculated using meq/L concentrations (Fig. 5f) to evaluate the direction of potential ion-exchange reactions [10]. Positive CAI values occurred in 15 wells (75%), whereas negative values occurred in 5 wells (25%), namely W4, W9, W10, W11, and W12. Positive CAI values are consistent with reverse cation exchange, whereas negative values are consistent with direct cation exchange, indicating spatially variable ion-exchange behavior across the aquifer. These additional diagnostics complement the existing ionic relationships and provide additional supporting evidence for a spatially variable contribution of cation exchange to groundwater evolution. Overall, the combined hydrochemical evidence indicates that water–rock interaction, particularly weathering of Ca- and Mg-bearing volcanic silicates, provides the better-supported explanation for the dominant groundwater chemistry, while carbonate-bearing phases remain a possible but unresolved contribution.

3.3.3 Correlation analysis of hydrochemical parameters

Pearson correlation analysis (Table 3) was performed to quantify the relationships among the major hydrochemical variables and to support the interpretation of the dominant geochemical processes governing groundwater chemistry.

Pearson correlation analysis revealed a near-perfect positive correlation between EC and TDS, confirming that electrical conductivity serves as a reliable indicator of total dissolved solids and overall groundwater mineralization. Strong positive correlations were also observed among EC, TDS, Ca^{2+} , Mg^{2+} , and HCO_3^- , indicating that groundwater chemistry is predominantly controlled by water–rock interaction processes. The strong association between Ca^{2+} and HCO_3^- is compatible with contributions from carbonate-bearing phases and weathering of Ca-bearing volcanic minerals; however, the available data do not permit their relative contributions to be quantified. The positive relationship with Mg^{2+} provides additional support for weathering of Mg-bearing volcanic silicates. Taken together, the hydrochemical facies, ionic relationships, and PHREEQC results indicate that water–rock interaction, particularly weathering of Ca- and Mg-bearing volcanic silicates, is the better-supported interpretation, while carbonate-bearing phases may provide a secondary but unresolved contribution.

In contrast, NO_3^- exhibited weak to moderate correlations with the major geogenic ions, suggesting that its spatial distribution is largely independent of the principal water–rock interaction processes. This pattern suggests localized anthropogenic influences rather than lithological controls, although the limited sample size warrants cautious interpretation. Overall, the correlation analysis supports the predominance of natural

Table 3 Pearson correlation coefficients among physicochemical parameters of groundwater samples

	pH	EC	TDS	Na ⁺	Ca ²⁺	Mg ²⁺	K ⁺	C ⁻	HCO ₃ ⁻	SO ₄ ²⁻	NO ₃ ⁻	F ⁻
pH	1.00											
EC	-0.65**	1.00										
TDS	-0.65**	1.00**	1.00									
Na ⁺	0.43	-0.28	-0.27	1.00								
Ca ²⁺	-0.82**	0.71**	0.71**	-0.63**	1.00							
Mg ²⁺	-0.24	0.69**	0.69**	-0.25	0.35	1.00						
K ⁺	-0.01	0.19	0.19	-0.01	0.10	0.34	1.00					
Cl ⁻	-0.57**	0.68**	0.68**	0.13	0.51*	0.39	0.09	1.00				
HCO ₃ ⁻	-0.69**	0.82**	0.82**	-0.31	0.77**	0.64**	0.19	0.50*	1.00			
SO ₄ ²⁻	-0.32	0.69**	0.69**	-0.07	0.44*	0.48*	0.17	0.60**	0.39	1.00		
NO ₃ ⁻	0.22	0.08	0.08	0.36	-0.13	0.45*	0.24	0.25	0.09	0.31	1.00	
F ⁻	0.41	-0.21	-0.21	0.30	-0.48*	-0.12	-0.18	-0.43	-0.18	-0.17	0.05	1.00

*Correlation is significant at the 0.05 level (two-tailed). **Correlation is significant at the 0.01 level (two-tailed)

Table 4 Saturation indices (SI) of selected mineral phases in groundwater samples using PHREEQC modeling

Well No.	Halite	Gypsum	Calcite	Aragonite	Anhydrite
W1	− 2.26	0.28	0.42	− 1.92	− 7.06
W2	− 1.98	1.02	1.16	− 1.67	− 6.93
W3	− 2.22	0.57	0.71	− 1.92	− 7.09
W4	− 2.79	0.16	0.30	− 2.45	− 6.38
W5	− 2.14	0.17	0.31	− 1.82	− 6.79
W6	− 1.98	0.31	0.45	− 1.67	− 6.65
W7	− 1.86	0.30	0.44	− 1.55	− 7.14
W8	− 2.07	0.22	0.37	− 1.76	− 7.00
W9	− 2.24	0.96	1.11	− 1.90	− 6.61
W10	− 2.39	0.41	0.55	− 2.07	− 6.68
W11	− 2.36	0.41	0.55	− 2.02	− 6.58
W12	− 2.91	− 0.03	0.12	− 2.60	− 6.94
W13	− 1.85	0.43	0.57	− 1.54	− 6.98
W14	− 1.93	0.56	0.71	− 1.62	− 6.97
W15	− 3.07	0.66	0.80	− 2.75	− 7.10
W16	− 2.32	1.03	1.17	− 2.01	− 7.01
W17	− 1.85	0.43	0.57	− 1.50	− 6.94
W18	− 2.04	0.37	0.51	− 1.73	− 6.71
W19	− 1.91	0.64	0.78	− 1.58	− 6.83
W20	− 2.35	0.41	0.56	− 2.03	− 7.24
Mean	− 2.23	0.47	0.61	− 1.91	− 6.88

An SI value greater than zero indicates oversaturation (precipitation tendency), $SI \approx 0$ denotes equilibrium conditions, and SI less than zero reflects undersaturation (dissolution tendency)

water–rock interaction processes in controlling groundwater chemistry while indicating that nitrate behaves relatively independently of the principal geogenic constituents.

3.3.4 Integrated interpretation

The combined evidence from Gibbs modeling, ionic relationships, and statistical correlations indicates that groundwater chemistry in the Ibb volcanic aquifer is predominantly governed by rock–water interaction within the volcanic lithologies, with weathering of Ca- and Mg-bearing volcanic silicates providing the better-supported explanation for the observed hydrochemical composition. Carbonate-bearing phases may contribute to Ca^{2+} and HCO_3^- , but their specific occurrence and relative contribution could not be established from the available data. Evaporation and evaporite dissolution exert negligible regional influence, whereas anthropogenic impacts are largely confined to localized nitrate enrichment. Overall, the results are consistent with lithological and water–rock interaction controls on hydrochemical evolution under recharge-dominated conditions.

3.4 Mineral saturation indices and water–rock interaction

Mineral saturation indices (SI) were calculated using PHREEQC to evaluate thermodynamic equilibrium between groundwater and selected mineral phases and to further constrain geochemical controls on aquifer evolution. Table 4 presents SI values for halite, gypsum, anhydrite, calcite, and aragonite. Positive SI values indicate oversaturation (precipitation tendency), $SI \approx 0$ reflects equilibrium, and negative SI values indicate undersaturation (dissolution tendency).

3.4.1 Carbonate mineral equilibrium

Groundwater samples were generally near equilibrium to moderately oversaturated with respect to carbonate minerals. Calcite saturation indices ranged from 0.12 to 1.17, with a mean of 0.61, whereas aragonite saturation indices ranged from -0.03 to 1.03, with a mean of 0.47. Most samples exhibited positive SI values for both carbonate minerals, indicating near-equilibrium to oversaturated conditions. W12 showed the lowest aragonite SI (-0.03), indicating near-equilibrium conditions, whereas W16 exhibited the highest calcite SI (1.17). The consistent occurrence of higher calcite SI values than aragonite SI values across the samples supports internal consistency of the revised PHREEQC calculations. However, saturation indices alone do not demonstrate active carbonate dissolution or establish carbonate minerals as the dominant source of Ca^{2+} and HCO_3^- . Therefore, the PHREEQC results are interpreted primarily as evidence of carbonate mineral equilibrium conditions rather than direct proof of a dominant carbonate dissolution mechanism.

3.4.2 Evaporite and sulfate mineral undersaturation

All groundwater samples exhibited strong undersaturation with respect to halite, gypsum, and anhydrite. Halite SI values ranged from -3.07 to -1.85 , with a mean of -2.23 ; gypsum ranged from -0.03 to 1.03, with a mean of 0.47; and anhydrite ranged from -7.24 to -6.38 , with a mean of -6.88 . The consistently negative SI values indicate that these mineral phases were thermodynamically undersaturated throughout the sampled groundwater system and were therefore unlikely to represent dominant equilibrium controls on groundwater mineralization at the time of sampling. The observed chloride and sulfate concentrations may instead reflect contributions from water–rock interaction, lithological sources, or other hydrochemical inputs that were not resolved directly by the saturation-index analysis.

3.4.3 Geochemical implications for aquifer evolution

The coexistence of near-equilibrium to moderately oversaturated conditions with respect to carbonate minerals and strong undersaturation with respect to halite, gypsum, and anhydrite indicates that groundwater mineralization was not predominantly controlled by evaporite mineral equilibrium. The carbonate saturation pattern is consistent with carbonate buffering and water–rock interaction under the sampled hydrochemical conditions, while the persistent undersaturation of evaporite and sulfate minerals indicates limited thermodynamic control by these phases. These results, together with the hydrochemical facies and ionic relationships, support the interpretation that groundwater chemistry reflects coupled water–rock interaction processes within the fractured volcanic aquifer. However, the relative contributions of individual mineral-weathering processes cannot be established solely from saturation indices.

3.5 Integrated drinking water quality assessment

Groundwater suitability for drinking purposes in the Ibb volcanic aquifer was evaluated using an integrated framework that combined physicochemical assessment based on the Water Quality Index (WQI) with microbiological assessment based on fecal coliform (FC) detection. Similar integrated groundwater quality assessment approaches have been widely applied in recent hydrogeochemical studies [17].

3.5.1 Physicochemical suitability (WQI)

The Water Quality Index (WQI) was calculated using eleven principal physicochemical parameters (pH, TDS, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , HCO_3^- , SO_4^{2-} , NO_3^- , and F^-), with World Health Organization guideline values adopted as reference standards. The computed WQI values ranged from 38 to 95 (Table 5), indicating overall moderate spatial variability in groundwater quality across the study area. Comparable groundwater-quality assessments have applied WQI together with hydrochemical and nitrate-related indicators to evaluate drinking-water suitability and potential health concerns [4, 9, 20, 21].

Based on the WQI classification proposed by Brown et al., [11], three wells (W10, W12, and W20) were classified as excellent water quality ($\text{WQI} < 50$), whereas the remaining 17 wells were classified as good quality (50–100). No groundwater samples fell into the poor, very poor, or unsuitable categories when evaluated solely on physicochemical parameters. Slightly elevated WQI values observed in wells W17 (95) and W9 (91) were primarily associated with increased total hardness and localized nitrate enrichment, reflecting intensified water–rock interaction and minor anthropogenic influence. Overall, the physicochemical profile indicated moderate mineralization consistent with carbonate–silicate weathering processes within the fractured volcanic aquifer. From a physicochemical perspective, groundwater quality was generally suitable for drinking purposes and complied with the applicable guideline thresholds.

3.5.2 Microbiological quality and public health implications

Fecal-coliform detection in five of the twenty sampled wells indicates localized sanitary concerns within the study area. Both WHO and Yemeni drinking-water standards require the absence of fecal indicator organisms in potable water; therefore, the FC-positive wells should not be considered suitable for direct human consumption without appropriate treatment. The occurrence of FC despite generally favorable physicochemical characteristics demonstrates that chemical compliance alone does not ensure microbiological safety. In fractured volcanic aquifers, interconnected fractures and joints may provide preferential pathways for groundwater movement and potentially facilitate microbial transport while reducing opportunities for natural filtration and attenuation [18]. However, fracture connectivity, contaminant travel times, source locations, and transport pathways were not directly investigated in the present study. Therefore, preferential flow and reduced attenuation should be regarded as plausible hydrogeological

Table 5 Water Quality Index (WQI) values and groundwater quality classification

Sample	WQI	Water quality class	Sample	WQI	Water quality class
W1	64	Good	W11	63	Good
W2	79	Good	W12	39	Excellent
W3	65	Good	W13	79	Good
W4	55	Good	W14	67	Good
W5	62	Good	W15	70	Good
W6	79	Good	W16	73	Good
W7	76	Good	W17	95	Good
W8	77	Good	W18	76	Good
W9	91	Good	W19	88	Good
W10	44	Excellent	W20	38	Excellent

The WQI results indicate generally favorable physicochemical conditions throughout the study area, with only minor spatial variability among the investigated wells

Table 6 Comparison of selected physicochemical parameters between FC-positive and FC-negative groundwater samples using the Mann–Whitney U test

Variable	FC-negative (median)	FC-positive (median)	Mann– Whitney U	Two-sided asymptotic <i>p</i> -value	Rank-biserial correlation	95% CI for rank-biserial correlation
WQI	70	77	44.0	0.600	– 0.173	– 0.707 to 0.360
EC (μS/cm)	946	1102	37.0	1.000	0.013	– 0.600 to 0.627
TDS (mg/L)	615	716	37.0	1.000	0.013	– 0.600 to 0.627
NO ₃ [–] (mg/L)	9.5	7.7	32.5	0.694	0.133	– 0.387 to 0.653

explanations rather than demonstrated mechanisms. The specific source and transport pathway of the detected fecal-coliform indicators were not established.

To examine whether fecal-coliform detection was associated with groundwater physicochemical characteristics, Mann–Whitney U tests were performed comparing FC-positive wells ($n = 5$) with FC-negative wells ($n = 15$). The statistical results are summarized in Table 6. No statistically significant differences were observed between the two groups for WQI, EC, TDS, or NO₃[–] (all $p > 0.05$). Rank-biserial correlation coefficients indicated negligible to weak effect sizes, suggesting that FC detection was not significantly associated with the investigated physicochemical parameters. Nevertheless, the small number of FC-positive wells ($n = 5$) limits the statistical power to detect small between-group differences.

3.5.3 Integrated safety perspective

The integrated assessment indicates that groundwater in the Ibb volcanic aquifer was generally favorable with respect to the measured physicochemical parameters, while fecal-coliform indicator detection identified localized sanitary concerns in a subset of wells. The absence of statistically significant differences in WQI, EC, TDS, and NO₃[–] between FC-positive and FC-negative wells indicates that the investigated physicochemical parameters did not distinguish the two groups in the present dataset. Accordingly, microbiological screening provides complementary information that is not captured by conventional physicochemical assessment alone. These findings support the inclusion of routine microbiological monitoring alongside hydrochemical assessment in groundwater-quality management.

3.6 Groundwater suitability for irrigation

Groundwater suitability for irrigation was evaluated using integrated salinity and sodicity indicators, including Electrical Conductivity (EC), Sodium Adsorption Ratio (SAR), Sodium Percentage (Na%), and Residual Sodium Carbonate (RSC). The calculated values are summarized in Table 7, and their combined relationships are illustrated in the USSS and Wilcox diagrams (Fig. 6). The calculated irrigation indices collectively indicate generally favorable irrigation water quality, although localized anomalies were identified in wells W4 and W12. The calculated irrigation indices collectively indicate generally favorable irrigation water quality, although localized anomalies were identified in wells W4 and W12.

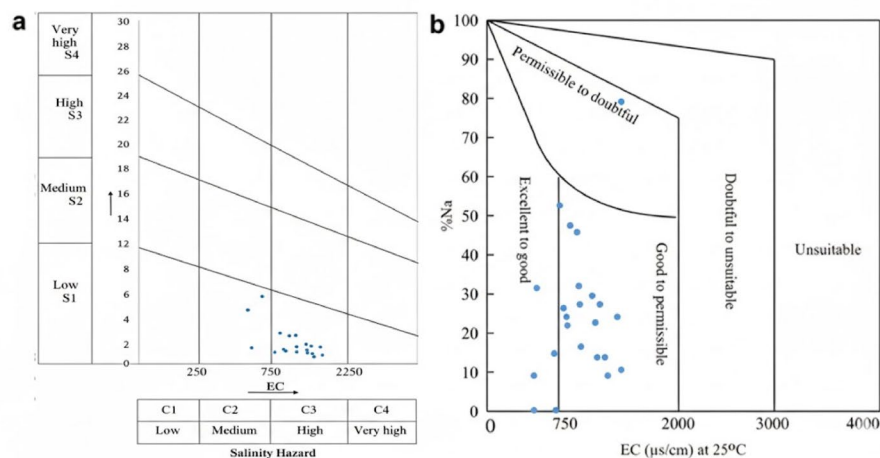
3.6.1 Salinity hazard (EC)

Electrical conductivity values ranged from 488 to 1406 μS/cm, indicating low to moderate salinity conditions across the sampled wells. According to the U.S. Salinity

Table 7 Irrigation water quality indices of groundwater samples

Well No.	SAR	Na%	RSC (meq/L)
W1	0.93	19.9	− 1.31
W2	0.98	18.9	− 2.29
W3	0.87	19.1	− 1.47
W4	6.46	77.5*	2.82*
W5	1.20	23.4	− 2.58
W6	1.47	26.6	− 1.72
W7	0.43	8.6	− 3.79
W8	0.77	14.7	− 3.12
W9	2.20	37.0	− 0.40
W10	2.33	46.1	− 1.42
W11	2.17	39.2	− 2.10
W12	4.38	76.5*	0.66
W13	0.73	13.6	− 3.52
W14	0.77	15.0	− 3.46
W15	1.02	22.5	0.52
W16	0.84	17.1	− 2.37
W17	0.57	10.2	− 4.53
W18	1.33	24.9	− 1.58
W19	1.20	22.2	− 1.46
W20	1.09	28.5	− 1.94
Min	0.43	8.6	− 4.53
Max	6.46	77.5	2.82
Mean	1.59	28.1	− 1.75
SD	1.47	20.4	1.95

* Exceeds recommended irrigation thresholds [31, 41]

**Fig. 6** Irrigation suitability classification of groundwater samples based on **a** the USSS diagram and **b** the Wilcox diagram

Laboratory (USSS) classification [31], the samples were distributed within the C2–C3 salinity classes. None of the wells exceeded the high-salinity range associated with the “doubtful” or “unsuitable” categories. The distribution of samples on the USSS diagram (Fig. 6a) indicates that groundwater salinity generally remains within manageable limits for agricultural use. While C3 waters may require appropriate leaching and drainage practices to minimize salt accumulation, they can remain suitable for irrigation depending on crop tolerance, soil conditions, and management practices.

3.6.2 Sodium hazard (SAR and Na%)

Sodium Adsorption Ratio (SAR) values ranged from 0.43 to 6.46, with a mean value of 1.59. All samples fell within the S1 category ($SAR < 10$), indicating low sodium hazard according to USSL criteria [31]. The low SAR values reflect the dominance of Ca^{2+} and Mg^{2+} over sodium, which favors soil structural stability and permeability. Most groundwater samples plotted within the C2–S1 and C3–S1 fields of the USSL diagram (Fig. 6a), representing moderate salinity coupled with low sodium hazard. This hydrochemical combination suggests limited risk of sodium-induced soil dispersion under conventional irrigation practices.

Sodium percentage (Na%) exhibited greater variability, ranging from 8.6 to 77.5%. The majority of wells remain below the recommended 60% threshold for irrigation suitability [41], confirming acceptable sodium levels in most locations. However, wells W4 and W12 exceeded this limit, indicating localized sodium enrichment. On the Wilcox diagram (Fig. 6b), these wells plotted within or near the “Doubtful” category, whereas the remaining samples fell within the “Excellent to Good” and “Good to Permissible” classes. Overall, sodium-related risks were spatially confined and did not characterize the aquifer as a whole.

3.6.3 Carbonate hazard (RSC)

Residual Sodium Carbonate (RSC) values ranged from -4.53 to 2.82 meq/L, with a mean value of -1.75 meq/L. Most samples exhibited negative RSC values, indicating an excess of calcium and magnesium relative to carbonate species. These conditions minimize the risk of carbonate precipitation and the subsequent development of soil sodicity [15]. Only well W4 exceeded the critical RSC threshold of 2.5 meq/L, classifying it as potentially unsuitable for long-term irrigation without corrective measures. Elevated RSC may promote precipitation of divalent cations, increasing the relative proportion of sodium at soil exchange sites, potentially leading to soil alkalinization and reduced permeability. However, this condition is localized and does not represent the overall aquifer system.

3.6.4 Overall irrigation suitability

The integrated evaluation of EC, SAR, Na%, and RSC (Table 7; Fig. 6) indicates that groundwater in the Ibb volcanic aquifer is generally suitable for irrigation use. Salinity levels are moderate and manageable, with sodium hazard consistently low based on SAR values. Carbonate-induced alkalinity risks are minimal, except for isolated anomalies. This favorable irrigation water quality reflects recharge-dominated hydrochemical conditions controlled by natural water–rock interactions, rather than by evaporative concentration or saline intrusion.

Although localized sodium enrichment is observed in wells W4 and W12, the overall irrigation suitability of the aquifer remains high. Periodic monitoring of salinity and sodium-related parameters is recommended to ensure sustainable soil management, especially under increasing groundwater abstraction and urban expansion pressures.

Among the analyzed wells, W4 showed the most distinct hydrochemical behavior, characterized by elevated sodium concentration (140 mg/L), relatively low calcium and magnesium concentrations, and high Na% and RSC values. This pattern may reflect localized cation-exchange and/or mixing processes. The simultaneous detection of

fecal-coliform indicators identifies W4 as a localized sanitary concern; however, the present study did not directly investigate fracture connectivity, contaminant travel times, or contamination sources. Therefore, no specific fracture-mediated transport mechanism can be established. Nevertheless, the combination of hydrochemical anomalies and fecal-coliform detection warrants priority groundwater-quality monitoring, sanitary inspection, and wellhead protection at this well.

3.7 Groundwater vulnerability and management implications

The integrated hydrogeochemical, microbiological, and irrigation assessments provide a comprehensive evaluation of groundwater quality and potential vulnerability in the Ibb volcanic aquifer. Although physicochemical parameters indicate generally acceptable quality for drinking and irrigation purposes, fecal-coliform indicator detection in five of the twenty sampled wells indicates localized sanitary concerns that are not apparent from physicochemical parameters alone.

The hydrochemical characteristics are predominantly governed by water–rock interaction under recharge-dominated conditions, with weathering of Ca- and Mg-bearing volcanic silicates providing the better-supported explanation for the observed mineralization and chemical buffering. Carbonate-bearing phases may contribute to Ca^{2+} and HCO_3^- , but their relative contribution could not be established from the available data. However, the fractured nature of the volcanic aquifer may facilitate preferential groundwater flow and rapid recharge under certain hydrogeological conditions, potentially reducing opportunities for natural filtration and attenuation. These processes were not directly quantified in the present study.

The observed difference between physicochemical suitability and fecal-coliform indicator detection highlights the importance of integrating microbiological screening into groundwater-quality monitoring programs. Potential sanitary sources in the study area include decentralized sanitation systems, septic tanks, and urban wastewater infrastructure; however, the present study did not establish the specific source, transport pathway, or persistence of the detected fecal-coliform indicators. Therefore, the observed indicator detection should not be interpreted as evidence of confirmed pathogen transport or a demonstrated causal relationship between specific sanitary sources and groundwater quality.

From a management perspective, wells showing fecal-coliform indicator detection should receive priority for sanitary inspection, wellhead protection, and appropriate treatment before use for drinking-water supply. Routine microbiological monitoring, together with hydrochemical monitoring, is recommended to identify changes in sanitary quality over time. Delineation of source-protection areas around production wells and improved management of septic systems and wastewater infrastructure may also reduce potential sanitary risks. At the urban scale, strengthening sanitation infrastructure, preventing wastewater leakage, and maintaining systematic groundwater-quality monitoring would support long-term protection of the aquifer. These measures should be considered precautionary management actions because the specific sources and transport pathways of the detected indicators were not established in the present study.

3.8 Comparison with other fractured volcanic aquifers

Groundwater systems hosted in fractured volcanic terrains in semi-arid regions commonly exhibit Ca–Mg–HCO₃ hydrochemical facies associated with recharge-driven water–rock interaction and weathering processes [3]. Similar structurally heterogeneous aquifers may exhibit preferential groundwater pathways that influence recharge, residence time, and natural attenuation processes [22, 24]. In such systems, preferential flow may reduce residence time and limit opportunities for natural attenuation of contaminants. In the present study, the detection of fecal-coliform indicators in five wells provides evidence of localized sanitary concerns; however, the specific microorganisms, sources, and transport pathways responsible for these detections were not established. Compared with fractured aquifers where progressive salinization or trace-element enrichment dominate water-quality concerns, the Ibb aquifer is characterized by moderate mineralization and limited evaporative influence. The principal groundwater-management concern identified in this study therefore relates to localized microbiological sanitary concerns and the need to integrate microbiological screening with conventional hydrochemical monitoring, while recognizing that structural heterogeneity may represent a potential contributing factor that requires further investigation. This comparison underscores the importance of considering structural hydrogeology as part of groundwater-quality and vulnerability assessment in fractured volcanic aquifers.

3.9 Study limitations

Although this study provides an integrated hydrogeochemical and microbiological assessment of groundwater in the fractured volcanic aquifer of Ibb City, several limitations should be acknowledged. First, groundwater sampling was conducted during the dry season, and therefore seasonal variations associated with recharge events were not evaluated. Seasonal changes in groundwater quality and fecal-coliform indicator detection may occur during periods of increased recharge and infiltration and should be investigated in future studies. Second, the number of sampled wells ($n = 20$) represents the principal municipal production wells supplying different sectors of Ibb City; however, this sample size may not fully capture the spatial heterogeneity of groundwater quality across the entire aquifer. Finally, the microbiological analysis focused on fecal coliform indicators as a screening measure of sanitary quality and did not include confirmatory testing for *Escherichia coli* or specific pathogenic microorganisms. Therefore, the identity, source, and transport pathway of the detected fecal-coliform indicators could not be established. Future investigations incorporating confirmatory microbiological methods, additional microbial indicators, and molecular techniques, together with sampling during different hydrological seasons, would improve characterization of sanitary conditions and potential groundwater vulnerability. Despite these limitations, the integrated hydrogeochemical–microbiological framework applied in this study provides a useful baseline assessment of groundwater quality and sanitary vulnerability in the Ibb volcanic aquifer.

4 Conclusions

This study presents an integrated hydrogeochemical and microbiological assessment of groundwater in the fractured volcanic aquifer of Ibb City, Yemen. The results indicate that groundwater chemistry is predominantly influenced by recharge-driven water–rock

interaction processes within the basaltic formations of the Tertiary Yemen Volcanic Group, with evidence of carbonate buffering and contributions from silicate weathering. Hydrochemical facies analysis identified the Ca–Mg–HCO₃ water type as dominant, while PHREEQC saturation-index modeling indicated near-equilibrium to oversaturated conditions with respect to carbonate minerals and persistent undersaturation with respect to halite, gypsum, and anhydrite, supporting the predominance of natural geogenic processes in controlling groundwater mineralization rather than evaporative concentration or saline intrusion. From a physicochemical perspective, groundwater is generally suitable for both drinking and irrigation purposes. Water Quality Index (WQI) values classified all groundwater samples as excellent to good, whereas irrigation indices indicated low sodium hazard and acceptable salinity conditions for agricultural use. However, fecal-coliform indicators were detected in five of the twenty sampled wells (W4, W6, W8, W13, and W14), indicating localized sanitary concerns and demonstrating that satisfactory hydrochemical quality does not necessarily ensure microbiological safety. Because the microbiological analysis was based on fecal-coliform indicator detection without confirmatory testing for *Escherichia coli* or specific pathogenic microorganisms, the identity, source, and transport pathway of the detected indicators were not established. Accordingly, these findings should be interpreted as an indicator-based sanitary concern rather than confirmation of a specific pathogen or contamination pathway.

The integrated hydrochemical, thermodynamic, statistical, irrigation, and microbiological assessments demonstrate that groundwater-quality evaluation in fractured volcanic aquifers should not rely solely on physicochemical indicators. Integration of hydrochemical characterization with microbiological screening is therefore important for identifying potential sanitary concerns that may not be apparent from conventional chemical parameters alone. Wells showing fecal-coliform indicator detection should receive appropriate sanitary inspection, continued microbiological monitoring, and suitable treatment before direct use for drinking-water supply. Overall, the Ibb volcanic aquifer represents an important groundwater resource with generally favorable physicochemical characteristics. However, the detection of fecal-coliform indicators in a subset of wells highlights the need for precautionary groundwater protection and monitoring, while the fractured hydrogeological setting warrants further investigation as a potential factor influencing groundwater vulnerability. Future investigations incorporating confirmatory microbiological testing, additional microbial indicators, molecular methods, and sampling across different hydrological seasons would help clarify the identity, source, and temporal variability of the detected indicators and further characterize groundwater sanitary vulnerability.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

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Author contributions

G.N. and H.A. conceived and designed the study. G.N. and H.A. conducted field investigations and groundwater sampling. H.A. performed hydrochemical analysis and geochemical modeling. H.A. prepared the original manuscript draft. M.N., B.A., N.A., and A.A. contributed to data interpretation and manuscript review. All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study did not involve human participants, animals, or human biological materials. Groundwater samples were collected with permission from the relevant authorities and/or well owners, where required. Therefore, ethical approval was not required.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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